

Topic D - Fields

D.1 Gravitational fields (SL)

Guiding questions

How are the properties of a gravitational field quantified?

How does an understanding of gravitational fields allow for humans to explore the solar system?

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D.1 Gravitational fields (SL)

Understandings:

- Kepler's three laws of orbital motion ***
- Newton's universal law of gravitation as given by $F = Gm_1m_2/r^2$ for bodies treated as point masses
- conditions under which extended bodies can be treated as point masses
- that gravitational field strength g at a point is the force per unit mass experienced by a small point mass at that point as given by $g = F/m = GM/r^2$
- gravitational field lines.

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Data booklet reference:

$$F = G \frac{m_1 m_2}{r^2}$$

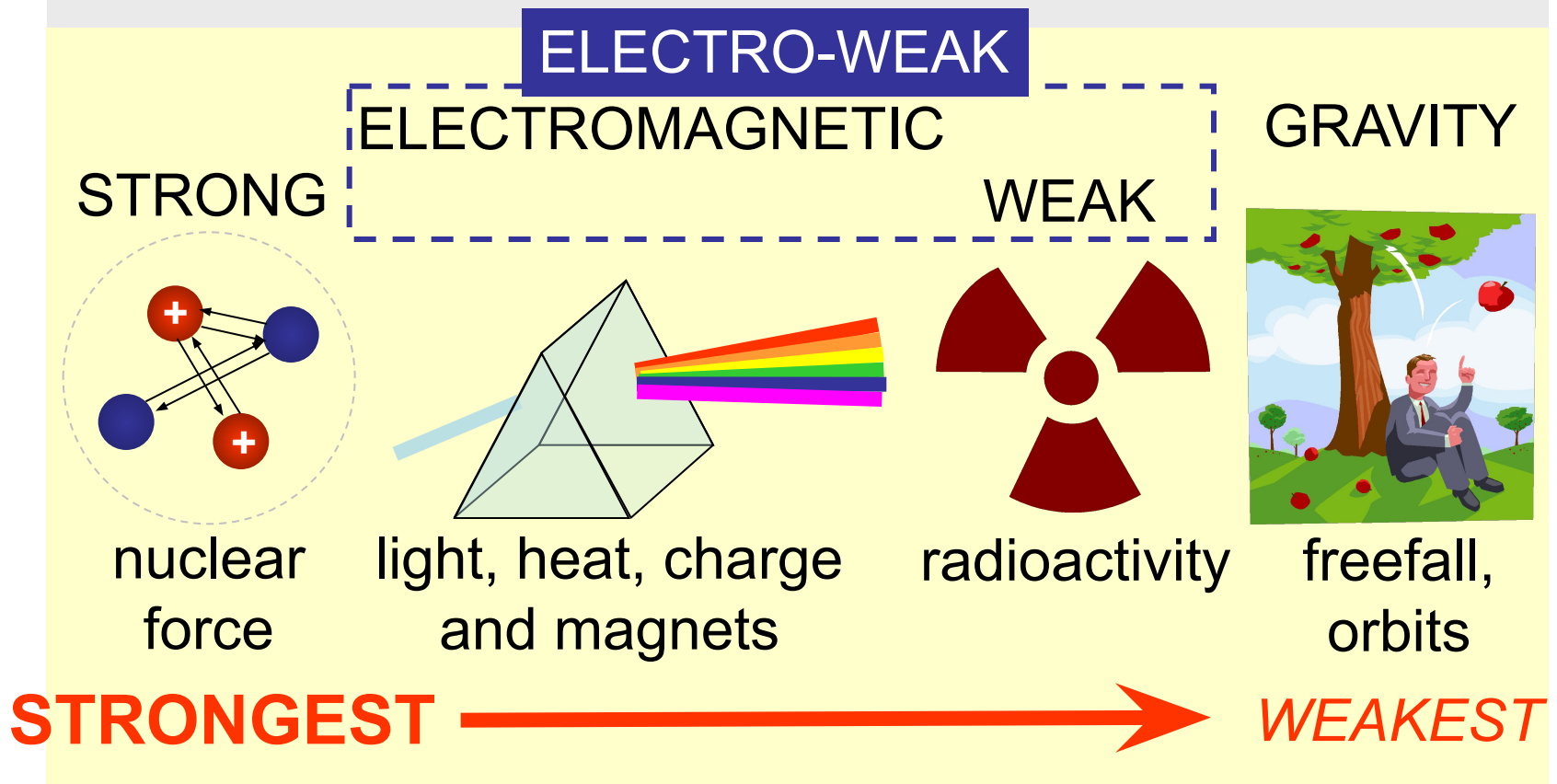
$$g = \frac{F}{m} = G \frac{M}{r^2}$$

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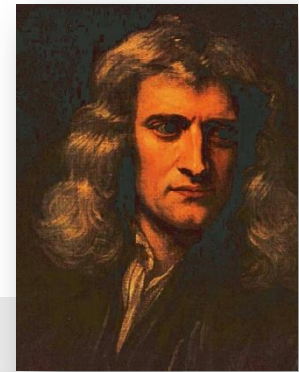
Newton's law of gravitation

·The gravitational force is the weakest of the four fundamental forces, as the following visual shows:



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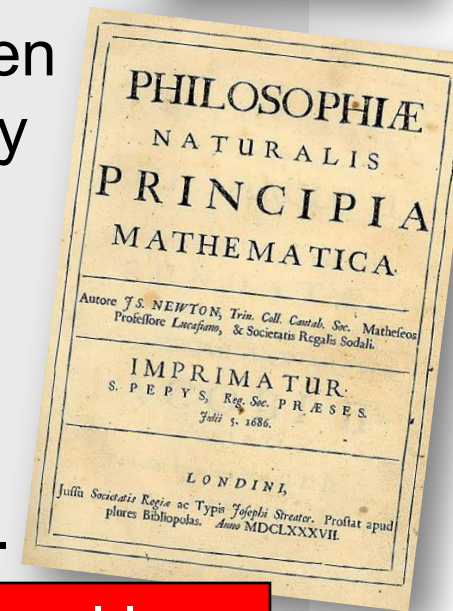
Newton's law of gravitation

- In 1687 Isaac Newton published what has been called by some the greatest scientific discovery of all time – his universal law of gravitation.
- The law states that the gravitational force between two **point masses** m_1 and m_2 is proportional to their product, and inversely proportional to the square of their separation r .

$$F = Gm_1m_2 / r^2$$

where $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

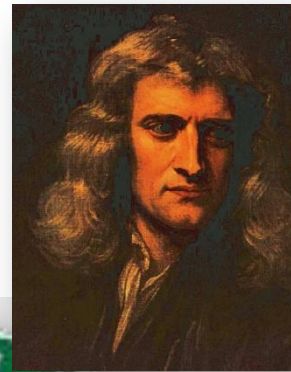
Universal law
of gravitation



- The actual value of G , the **universal gravitational constant**, was not known until Henry Cavendish conducted a tricky experiment in 1798 to find it.

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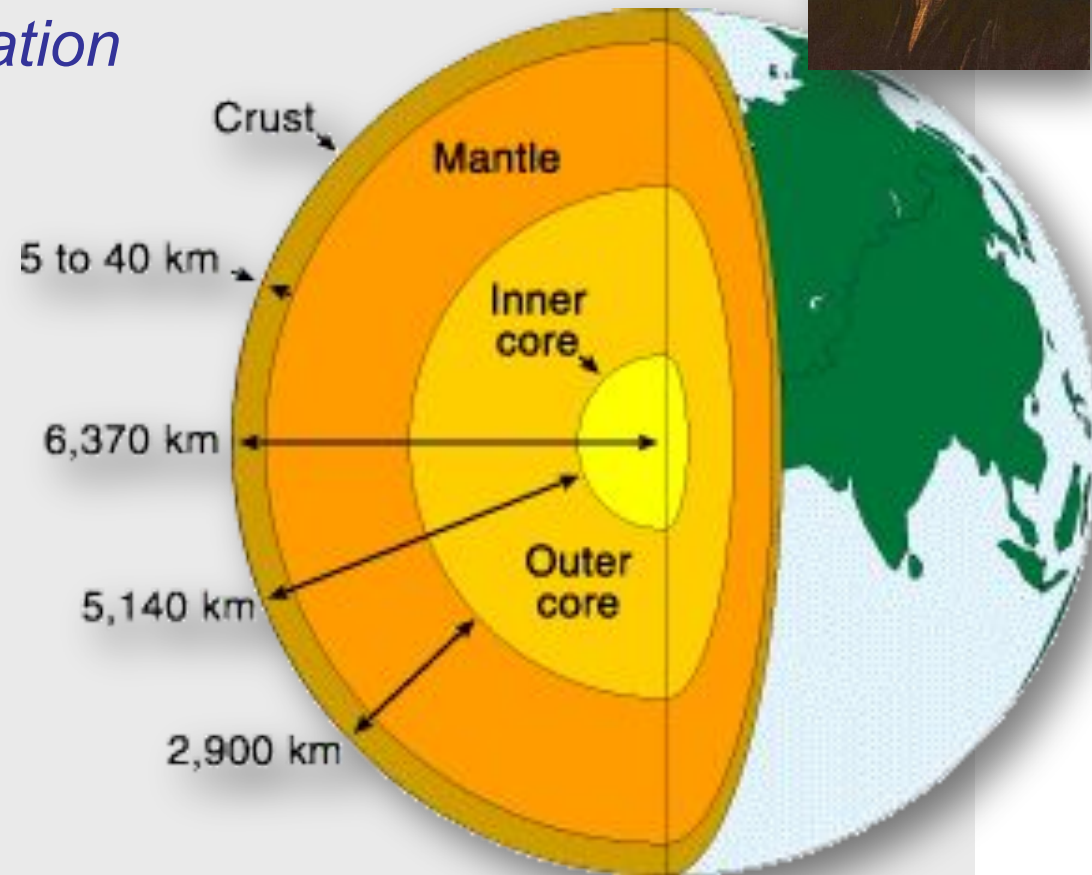


Newton's law of gravitation

·The Earth, planets, moons, and even the sun, have layers – kind of like an onion:

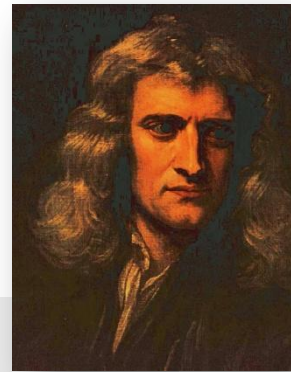
·In other words, **NONE** of the celestial bodies we observe are point masses.

·Given that the law is called the universal law of gravitation, how do we use it for planets and such?



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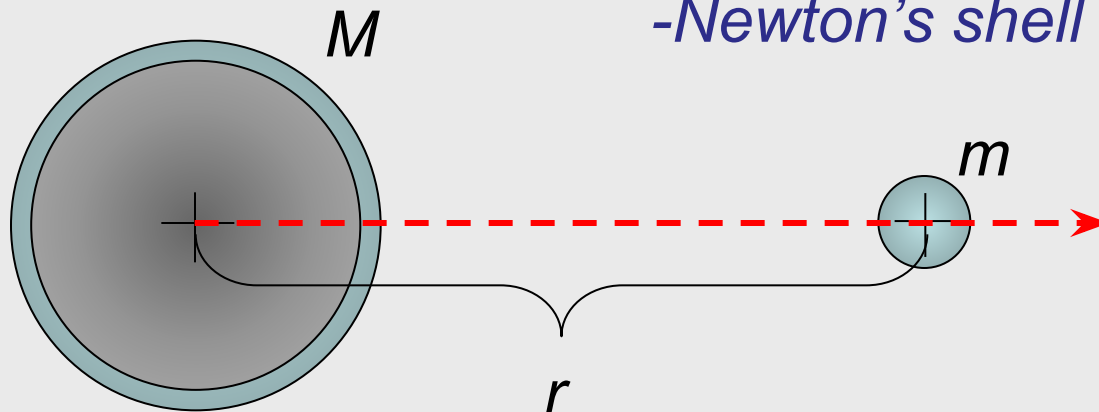


Newton's law of gravitation

·Newton spent much time developing integral calculus to prove that

“A spherically symmetric shell of mass M acts as if all of its mass is located at its center.”

-Newton's shell theorem.



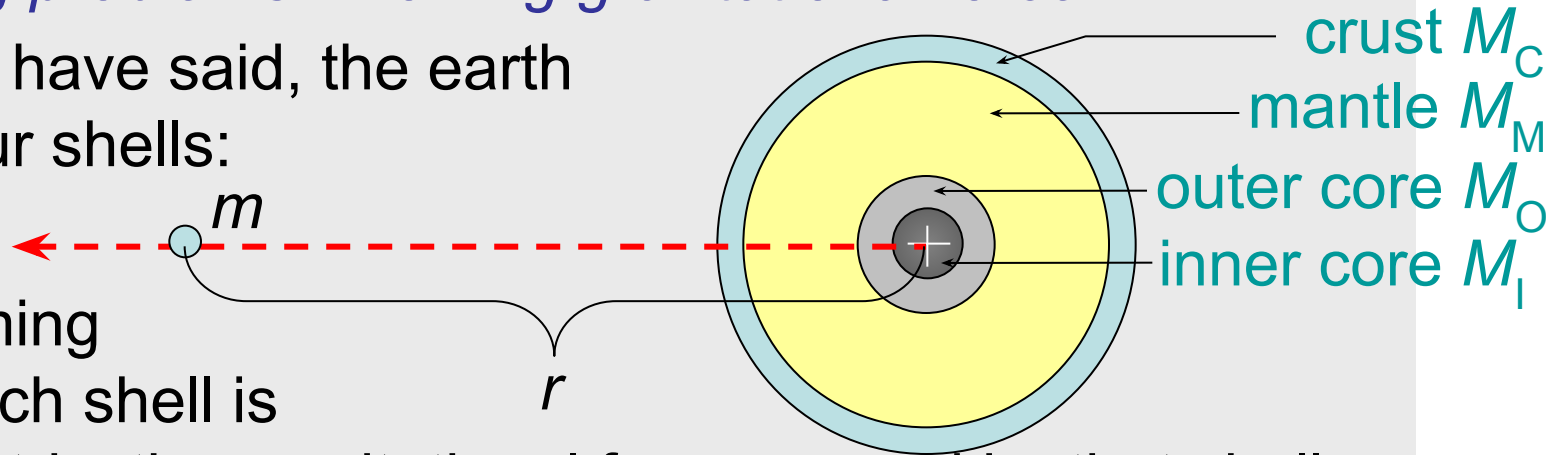
·Thus $F = Gm_1m_2 / r^2$ works not only for point masses, which have no radii, but for any spherically symmetric distribution of mass at any radius like planets and stars.

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Solving problems involving gravitational force

·As we have said, the earth has four shells:



·Assuming that each shell is symmetric, the gravitational force caused by that shell acts as though its mass is all concentrated at its center.

·The net force at m caused by the shells is given by

$$F = GM_I m / r^2 + GM_O m / r^2 + GM_M m / r^2 + GM_C m / r^2$$

$$F = G(M_I + M_O + M_M + M_C)m / r^2$$

·Thus $F = GMm / r^2$ where $M = M_I + M_O + M_M + M_C$.

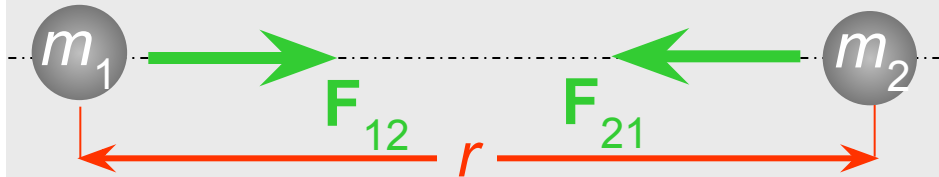
·Note that M is just the total mass of the earth.

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Solving problems involving gravitational force

·Be very clear that r is the **distance between the centers** of the masses.



FYI

·The radius of each mass is immaterial.

EXAMPLE: The earth has a mass of $M = 5.98 \times 10^{24}$ kg and the moon has a mass of $m = 7.36 \times 10^{22}$ kg. The mean distance between the earth and the moon is 3.82×10^8 m. What is the gravitational force between them?

SOLUTION: Use $F = GMm / r^2$.

$$F = (6.67 \times 10^{-11})(5.98 \times 10^{24})(7.36 \times 10^{22}) / (3.82 \times 10^8)^2$$

$$F = 2.01 \times 10^{20} \text{ N.}$$

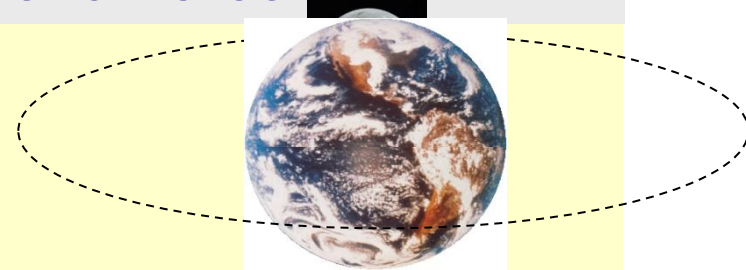


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Solving problems involving gravitational force

EXAMPLE: The moon has a mass of $m = 7.36 \times 10^{22}$ kg. The mean distance between the earth and the moon is 3.82×10^8 m. What is the speed of the moon in its orbit about earth?



SOLUTION: Use $F_C = F_G = mv^2 / r$.

• From the previous slide $F_G = 2.01 \times 10^{20}$ N. Then

$$2.01 \times 10^{20} = (7.36 \times 10^{22}) v^2 / 3.82 \times 10^8$$

• Then $v = 1.02 \times 10^3 \text{ ms}^{-1}$.

FYI

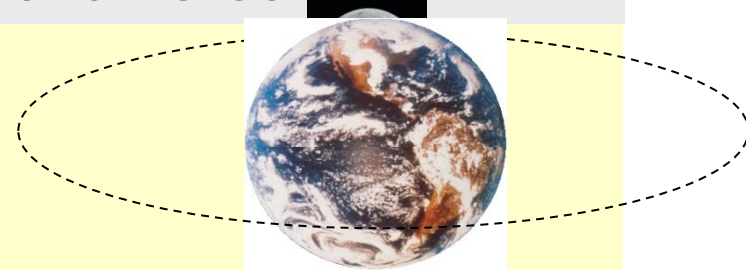
• For circular orbits, the gravitational force *is* the centripetal force. Thus $F_C = F_G$.

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Solving problems involving gravitational force

EXAMPLE: The moon has a mass of $m = 7.36 \times 10^{22}$ kg. The mean distance between the earth and the moon is 3.82×10^8 m. What is the period of the moon (in days) in its orbit about earth?



SOLUTION: Use $v = d / t = 2\pi r / T$.

• From the previous slide $v = 1.02 \times 10^3 \text{ ms}^{-1}$. Then

$$\begin{aligned} T &= 2\pi r / v = 2\pi(3.82 \times 10^8) / 1.02 \times 10^3 \\ &= (2.35 \times 10^6 \text{ s})(1 \text{ h} / 3600 \text{ s})(1 \text{ d} / 24 \text{ h}) = 27.2 \text{ d.} \end{aligned}$$

FYI

• For circular orbits, the gravitational force *is* the centripetal force. Thus $F_c = F_g$.

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D.1 Gravitational fields (SL)

Gravitational field strength

- Suppose a mass m is located a distance r from a another mass M .
- The **gravitational field strength** g is the force per unit mass acting on m due to the presence of M . Thus

$$g = F / m$$

gravitational field strength

- Question: What is the weight W of a 100 kg person standing on the surface of the Earth?
- Answer: $F_g = mg = (100 \text{ kg})(9.8 \text{ m s}^{-2}) = 980 \text{ N}$
- Question: What is the gravitational field strength g where that person stands?
- Answer: $g = (980 \text{ N}) / (100 \text{ kg}) = 9.8 \text{ N kg}^{-1} = 9.8 \text{ m s}^{-2}$.

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D.1 Gravitational fields (SL)

Gravitational field strength

- Suppose a mass m is located a distance r from a another mass M .
- The **gravitational field strength** g is the force per unit mass acting on m due to the presence of M . Thus

$$g = F / m$$

gravitational field strength

- The units are newtons per kilogram (N kg^{-1}).
- Note that from Newton's second law, $F = ma$, we see that a N kg^{-1} is also a m s^{-2} , the units for acceleration.
- Note further that weight has the formula $F_g = mg$, and that the g in this formula is none other than the gravitational field strength!
- On the earth's surface, $g = 9.8 \text{ N kg}^{-1} = 9.8 \text{ m s}^{-2}$.

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Solving problems involving gravitational field strength

- Suppose a mass m is located on the surface of a planet of radius R . We know that its weight is $F = mg$.
- But from the law of universal gravitation, the weight of m is equal to its attraction to the planet's mass M and equals $F = GMm / R^2$.
- Thus $mg = GMm / R^2$.

$$g = GM / R^2$$

gravitational field strength at surface of a planet of mass M and radius R

- This same derivation works for any m or r .

$$g = GM / r^2$$

gravitational field strength at distance r from center of a mass M

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D.1 Gravitational fields (SL)

Solving problems involving gravitational field strength

PRACTICE: The mass of the earth is $M = 5.98 \times 10^{24}$ kg and the radius of the earth is $R = 6.37 \times 10^6$ m. Find the gravitational field strength at the surface of the earth, and at a distance of one earth radius above its surface.

SOLUTION:

·For $r = R$:

$$g = GM / R^2$$

$$g = (6.67 \times 10^{-11})(5.98 \times 10^{24}) / (6.37 \times 10^6)^2$$

$$g = 9.83 \text{ N kg}^{-1} \text{ (m s}^{-2}\text{)}.$$

·For $r = 2R$: Since r is squared...just divide by $2^2 = 4$.

Thus

$$g = 9.83 / 4 = 2.46 \text{ m s}^{-2}.$$

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Solving problems involving gravitational field strength

PRACTICE: A 525-kg satellite is launched from the earth's surface to a height of one earth radius above the surface. What is its weight (a) at the surface, and (b) at altitude?

SOLUTION: Use information from the previous slide:

(a) AT SURFACE: $g_{\text{surface}} = 9.83 \text{ m s}^{-2}$.

·Then from $F = mg$ we get

$$F = (525)(9.83) = 5160 \text{ N.}$$

(b) AT ALTITUDE: $g_{\text{surface}+R} = 2.46 \text{ m s}^{-2}$.

·Then from $F = mg$ we get

$$F = (525)(2.46) = 1290 \text{ N.}$$

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Gravitational field strength

- Compare the gravitational force formula

$$F = GMm / r^2 \quad (\text{Force – action at a distance})$$

with the gravitational field formula

$$g = GM / r^2 \quad (\text{Field – local curvature of space})$$

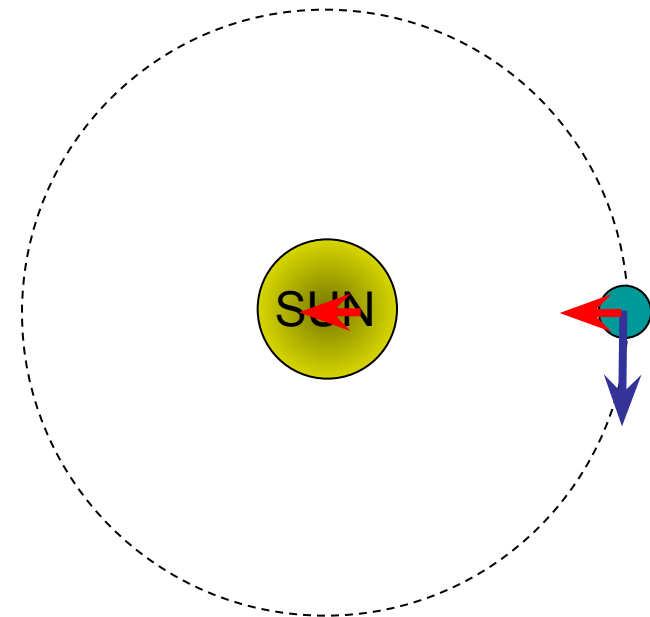
- Note that the force formula has **two masses**, and the force is the result of their interaction **at a distance r** .
- Note that the field formula has just **one mass** – namely the mass that “sets up” the local field in the space surrounding it. It “curves” it.
- The field view of the universe (**spatial disruption by a single mass**) is currently preferred over the force view (**action at a distance**) as the next slides will show.

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Gravitational field strength

- Consider the force view (action at a distance).
- In the force view, the masses know the locations of each other at all times, and the force is instantaneously felt by both masses at all times.
- This requires the “force signal” to be transferred between the masses instantaneously.
- As we will learn later, Einstein’s special theory of relativity states unequivocally that the fastest **any** signal can travel is at the (finite) speed of light c .

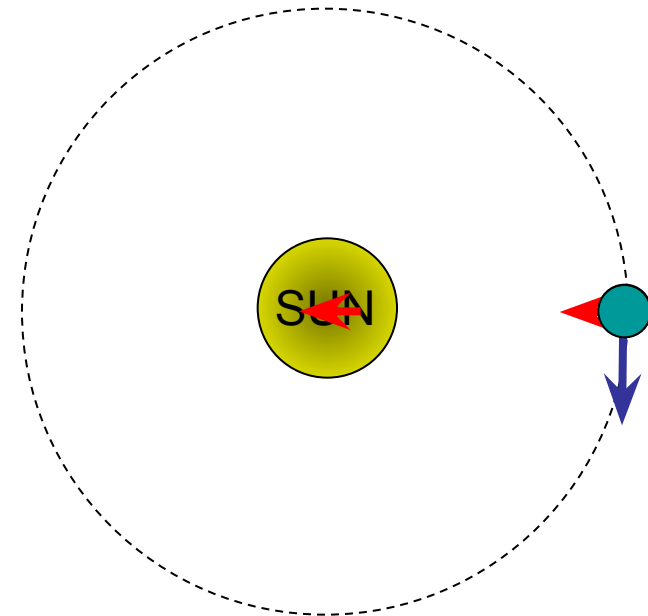


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Gravitational field strength

- Thus the action at a distance “force signal” will be slightly delayed in telling the orbital mass when to turn.
- The end result would have to be an expanding spiral motion, as illustrated in the following animation:
- We do **not** observe planets leaving their orbits as they travel around the sun.
- Thus action at a distance doesn't work if we are to believe special relativity.
- And all current evidence points to the correctness of special relativity.

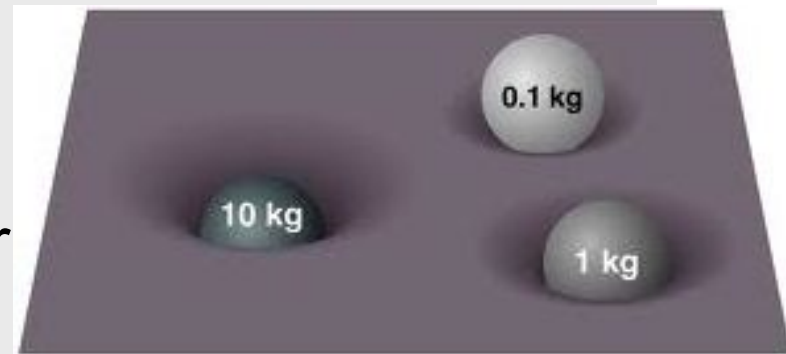


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Gravitational field strength

- So how does the field view take care of this “signal lag” problem?
- Simply put – the gravitational field distorts the space around the mass that is causing it so that any other mass placed at any position in the field will “know” how to respond immediately.
- Think of space as a stretched rubber sheet – like a drum head.
- Bigger masses “curve” the rubber sheet more than smaller masses.
- The next slide illustrates this gravitational “curvature” of the space surrounding, for example, the sun.

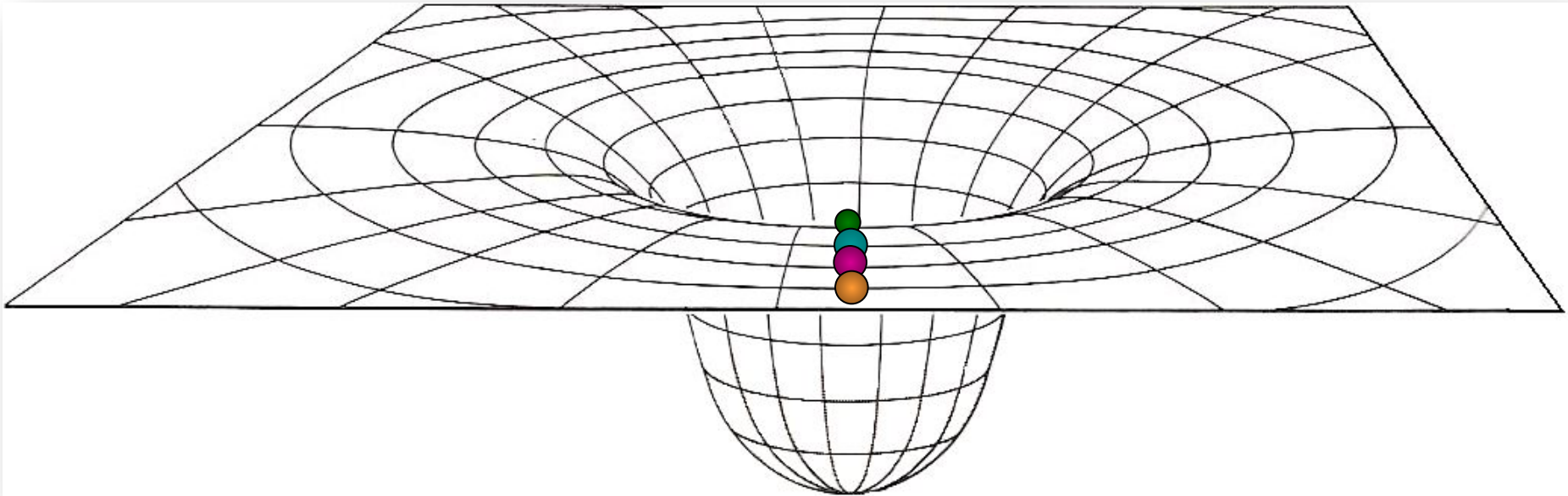


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Gravitational field strength

- Note that each mass “feels” a different “slope” and must travel at a particular speed to stay in orbit.



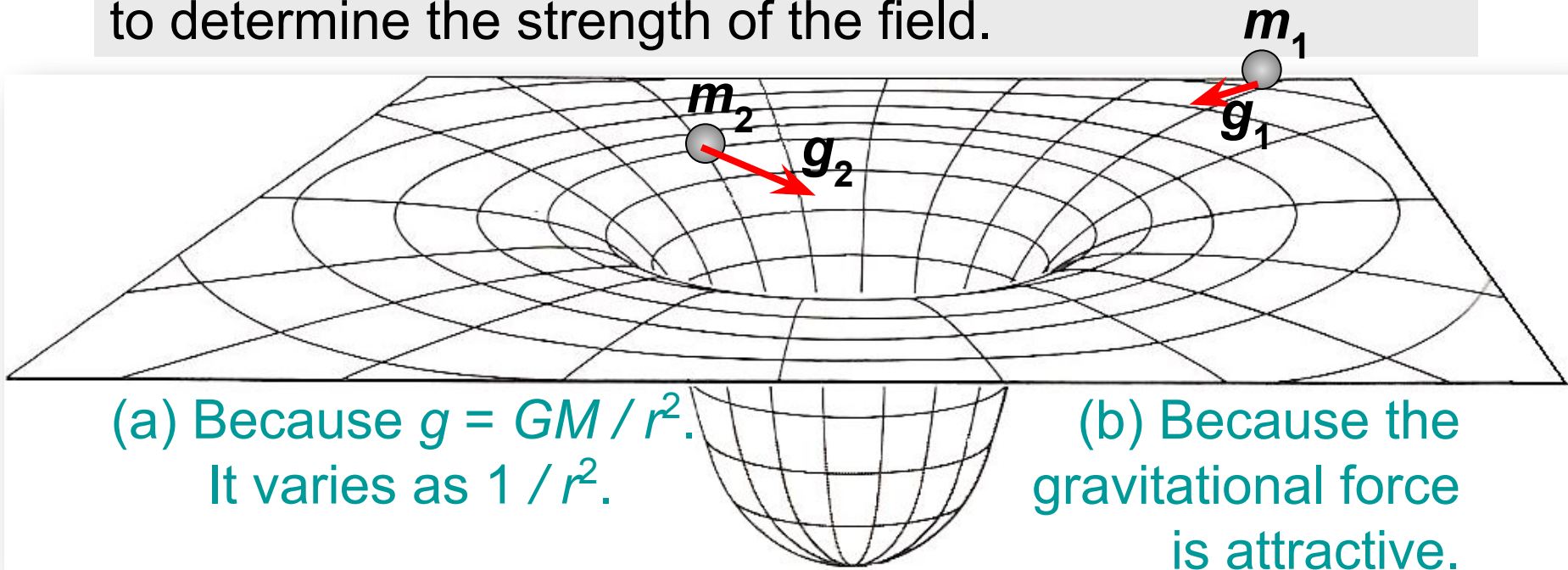
- The field view eliminates the need for long distance signaling between two masses. Rather, it distorts the space about one mass.

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D.1 Gravitational fields (SL)

Gravitational field strength

· In the space surrounding the mass M which sets up the field we can release “test masses” m_1 and m_2 as shown to determine the strength of the field.



(a) The field arrow is bigger for m_2 than m_1 . Why?

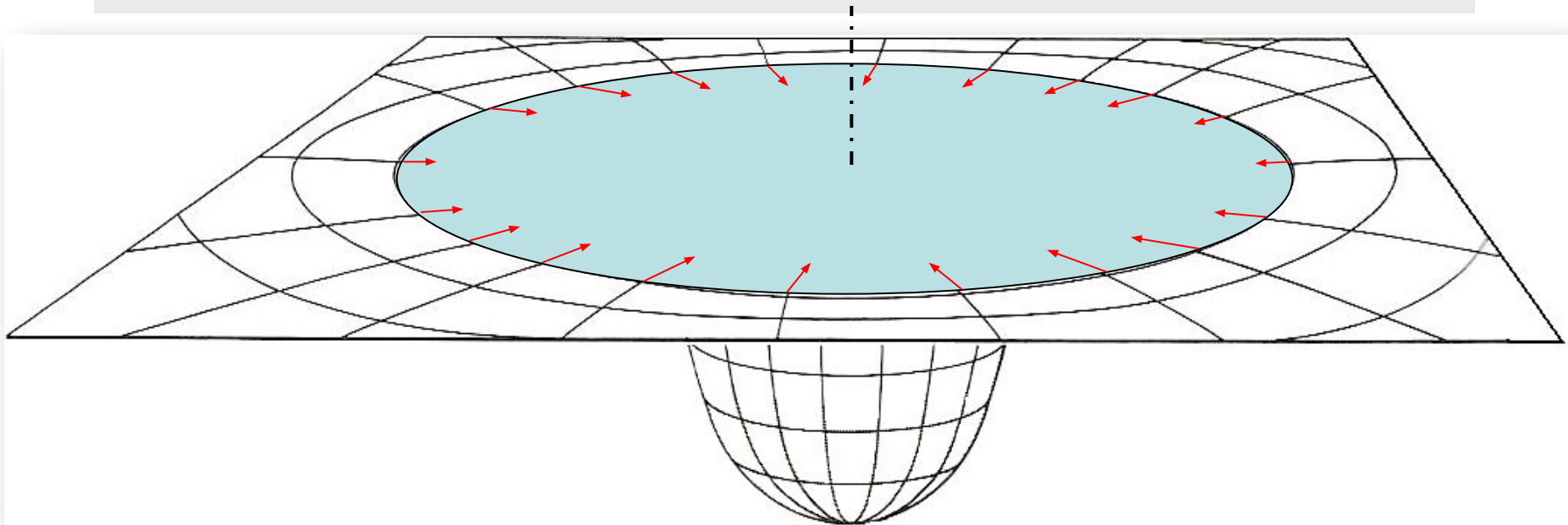
(b) The field arrow always points to M . Why?

Topic 6: Circular motion and gravitation

6.2 – Newton's law of gravitation

Gravitational field strength

·By “placing” a series of small **test masses** about a larger mass, we can map out its gravitational field:



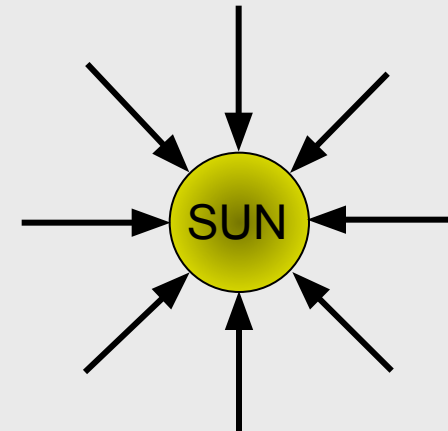
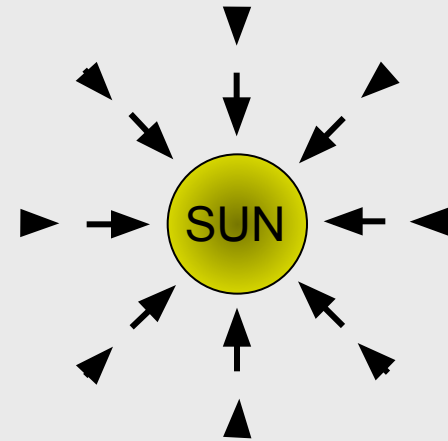
·The field arrows of the inner ring are longer than the field arrows of the outer ring and all field arrows point to the centerline.

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Gravitational field strength

- If we take a top view, and eliminate some of the field arrows, our sketch of the gravitational field is vastly simplified:
- In fact, we don't even have to draw the sun – the arrows are sufficient to denote its presence.
- To simplify field drawings even more, we take the convention of drawing “field lines” as a single arrow.

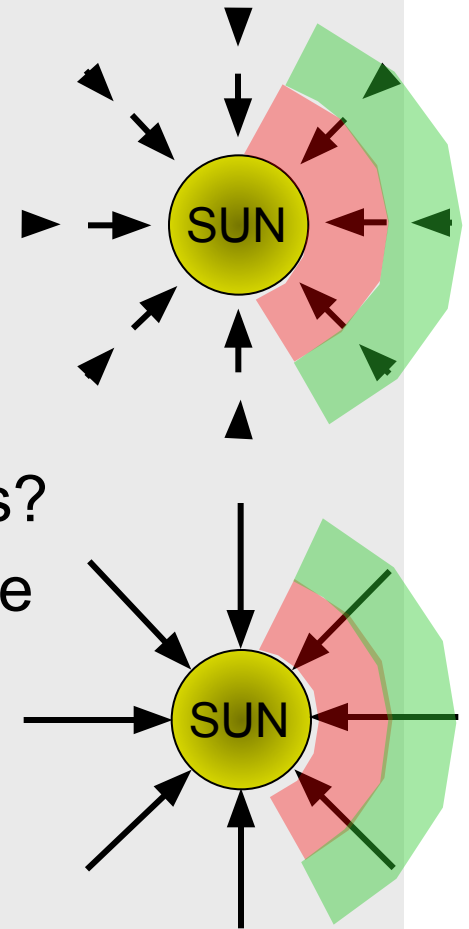


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D.1 Gravitational fields (SL)

Gravitational field strength

- In the first sketch the strength of the field at a point is determined by the length of the field arrows in the vicinity of that point.
- The second sketch has single arrows, so how do we know how strong the field is at a particular point in the vicinity of a mass?
- We simply look at the **concentration** of the field lines. *The closer together the field lines, the stronger the field.*
- In the **red region** the field lines are closer together than in the **green region**.
- Thus the **red field** is stronger than the **green field**.



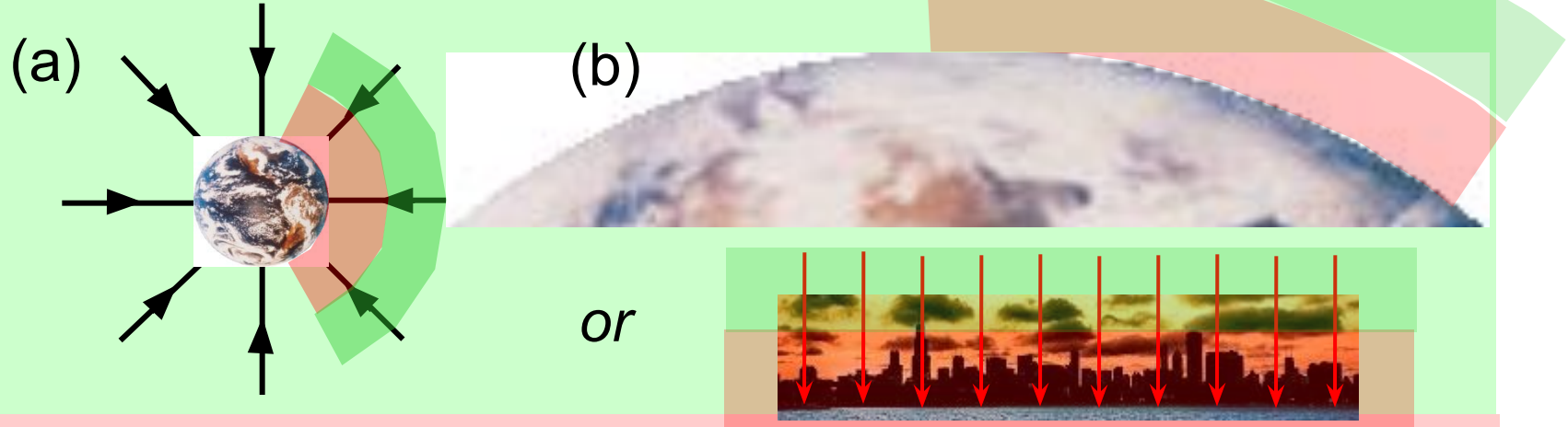
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D.1 Gravitational fields (SL)

Solving problems involving gravitational field strength

PRACTICE: Sketch the gravitational field about the earth (a) as viewed from far away, and (b) as viewed “locally” (at the surface).

SOLUTION:



FYI • Note that the closer to the surface we are, the more uniform the field concentration.

• At the surface, the field lines are approx. parallel!

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D.1 Gravitational fields (SL)

Solving problems involving gravitational field strength

EXAMPLE: Find the gravitational field strength at a point between the earth and the moon that is right between their centers.

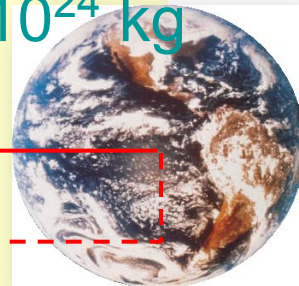
SOLUTION:

· Make a sketch.



$$m = 7.36 \times 10^{22} \text{ kg}$$

$$M = 5.98 \times 10^{24} \text{ kg}$$



g_m

g_M

$$d = 3.82 \times 10^8 \text{ m}$$

· Note that $r = d / 2 = 3.82 \times 10^8 / 2 = 1.91 \times 10^8 \text{ m}$.

$$g_m = Gm / r^2$$

$$g_m = (6.67 \times 10^{-11})(7.36 \times 10^{22}) / (1.91 \times 10^8)^2 = 1.35 \times 10^{-4} \text{ N kg}^{-1}$$

$$g_M = GM / r^2$$

$$g_M = (6.67 \times 10^{-11})(5.98 \times 10^{24}) / (1.91 \times 10^8)^2 = 1.09 \times 10^{-2} \text{ N kg}^{-1}$$

· Finally, $g = g_M - g_m = 1.08 \times 10^{-2} \text{ N kg}^{-1}, \rightarrow$

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D.1 Gravitational fields (SL)

Solving problems involving gravitational field strength

PRACTICE: Jupiter's gravitational field strength at its surface is 25 N kg^{-1} while its radius is $7.1 \times 10^7 \text{ m}$.

(a) Derive an expression for the gravitational field strength at the surface of a planet in terms of its mass M and radius R and the gravitational constant G .

SOLUTION: This is for a general planet...

(a) $F = Gm_1m_2 / r^2$ (law of universal gravitation)

$$F = GMm_2 / R^2 \quad (\text{substitution})$$

$$g = F / m_2 \quad (\text{gravitational field definition})$$

$$g = (GMm_2 / R^2) / m_2 \quad (\text{substitution})$$

$$g = GM / R^2$$

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D.1 Gravitational fields (SL)

Solving problems involving gravitational field strength

PRACTICE: Jupiter's gravitational field strength at its surface is 25 N kg^{-1} while its radius is $7.1 \times 10^7 \text{ m}$.

(b) Using the given information and the formula you just derived deduce Jupiter's mass.

(c) Find the weight of a 65-kg man on Jupiter.

SOLUTION:

(b) $g = GM / R^2$ (just derived in (a))

$$M = gR^2 / G \text{ (manipulation)}$$

$$M = (25)(7.1 \times 10^7)^2 / 6.67 \times 10^{-11} = 1.9 \times 10^{27} \text{ kg.}$$

(c) $F = mg$

$$F = 65(25) = 1600 \text{ N.}$$

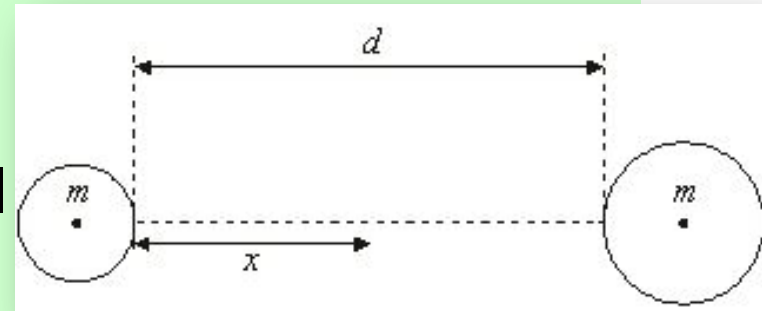


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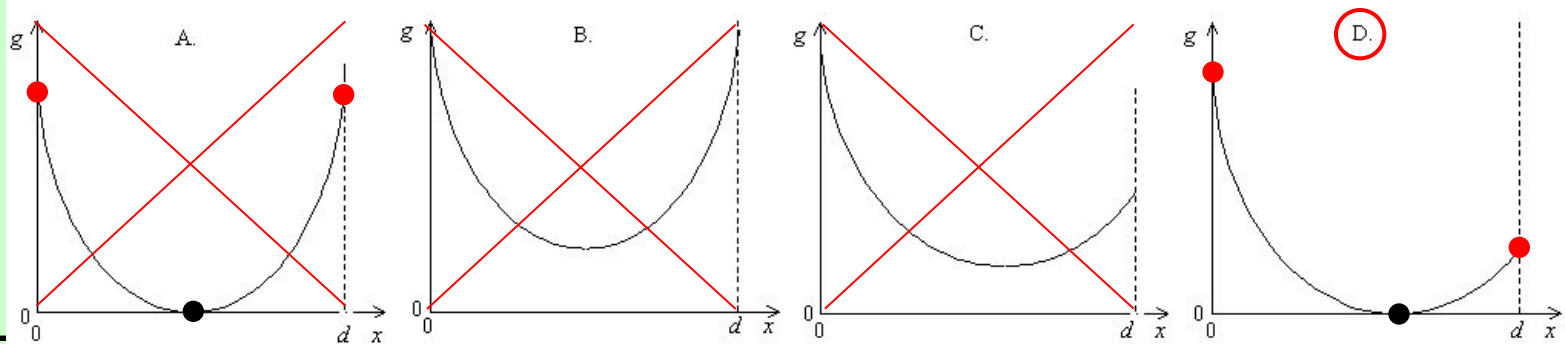
D.1 Gravitational fields (SL)

Solving problems involving gravitational field strength

PRACTICE: Two spheres of equal mass and different radii are held a distance d apart. The gravitational field strength is measured on the line joining the two masses at position x which varies.



Which graph shows the variation of g with x correctly?



• There is a point between M and m where $g = 0$.

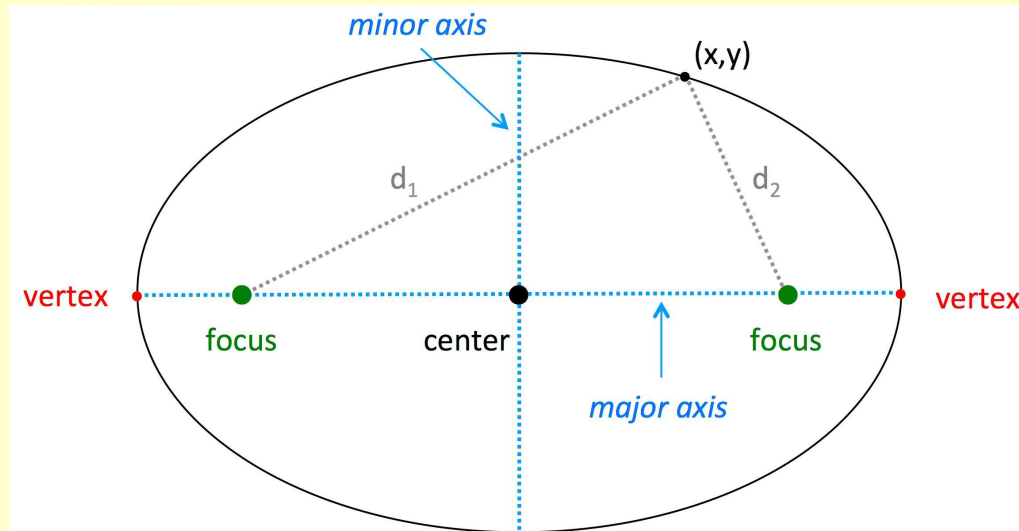
• Since $g = Gm / R^2$ and $R_{\text{left}} < R_{\text{right}}$, then $g_{\text{left}} > g_{\text{right}}$ at the surfaces of the masses.

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Kepler's Laws

In reality, orbits are not perfect circles. The shape of an orbit is actually an **ellipse**.



An ellipse is formed when a point (x,y) is traced around two focus points (called “foci”) where the sum of the distances between the point and each focus ($d_1 + d_2$) remains constant.

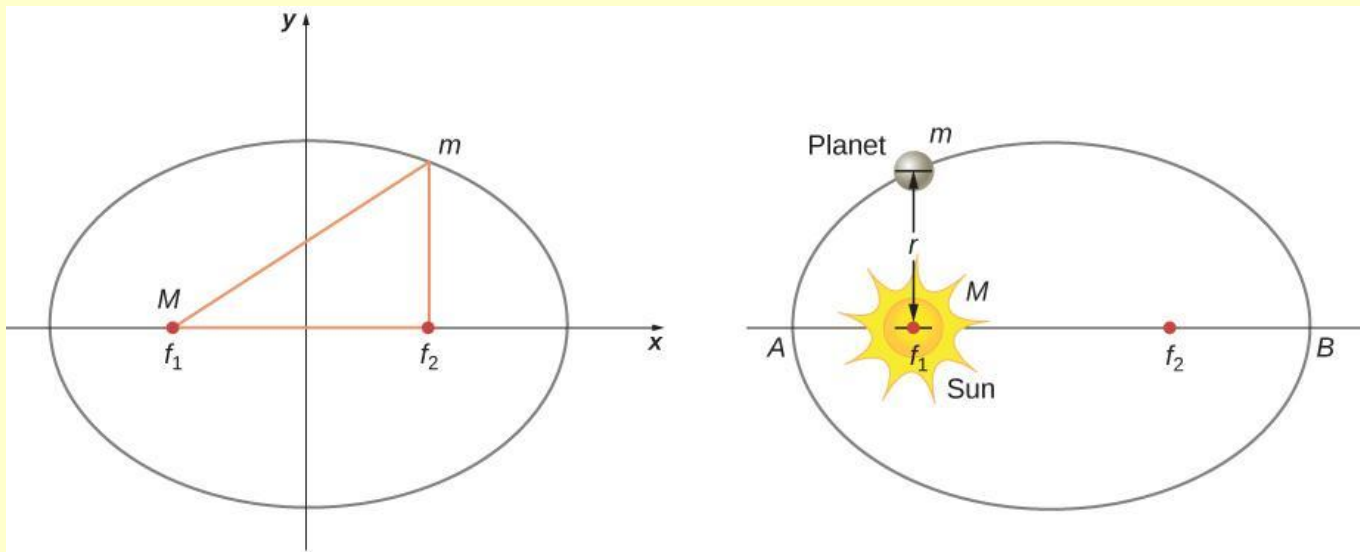
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Kepler's Laws

Johannes Kepler, a German astronomer and mathematician, developed 3 Laws of Planetary Motion that described how planets orbit stars.

Kepler's 1st Law: explains that planets are orbiting the sun in a path described as an ellipse, and the sun is located at one focus.



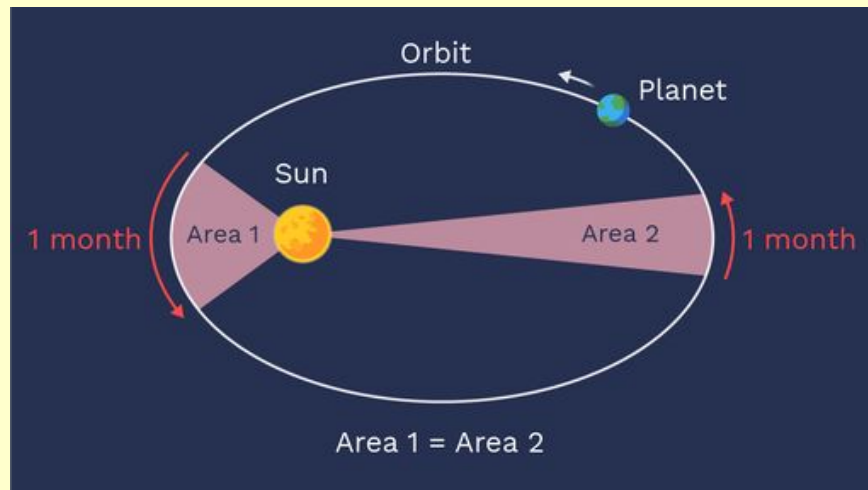
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Kepler's Laws



Kepler's 2nd Law: describes the speed at which any given planet will move while orbiting the sun. A planet moves *fastest* when it is *closest* to the sun and slowest when it is furthest from the sun. Yet, if an imaginary line were drawn from the center of the planet to the center of the sun, that line would sweep out the **same area in equal periods of time**.



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Kepler's Laws



EXAMPLE: Derive **Kepler's 3rd Law**, which states that the period T of an object in a circular orbit about a body of mass M is given by $T^2 = [(4\pi^2 / (GM))] r^3$.

SOLUTION:

- In circular orbit $F_c = ma_c$.
- From Newton's law of gravitation $F_c = GMm / r^2$.
- From Topic A.2, $a_c = 4\pi^2 r / T^2$. Then

$$\cancel{m}a_c = GM\cancel{m} / r^2$$

$$4\pi^2 r / T^2 = GM / r^2$$

$$4\pi^2 r^3 = GMT^2$$

$$T^2 = [4\pi^2 / (GM)] r^3.$$

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Solving problems involving orbital period

EXAMPLE: A satellite in geosynchronous orbit takes 24 hours to orbit the earth.

Thus, it can be above the same point of the earth's surface at all times, if desired.

Find the necessary orbital radius, and express it in terms of earth radii. $R_E = 6.37 \times 10^6 \text{ m}$.

SOLUTION: $T = (24 \text{ h})(3600 \text{ s h}^{-1}) = 86400 \text{ s}$.

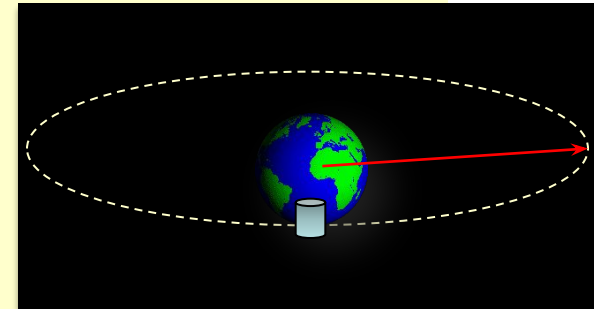
Then from Kepler's law $T^2 = [4\pi^2 / (GM)] r^3$ we have

$$r^3 = T^2 / [4\pi^2 / (GM)]$$

$$r^3 = 86400^2 / [4\pi^2 / (6.67 \times 10^{-11} \times 5.98 \times 10^{24})]$$

$$= 7.54 \times 10^{22}$$

$$r = (42250474 \text{ m})(1 R_E / 6.37 \times 10^6 \text{ m}) = 6.63 R_E.$$



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Solving problems involving orbital period

Which of the following expressions correctly relates the radius R of the circular orbit of a planet round a star to the period T of the orbit?

- A. $R^3 \propto T^2$ B. $\frac{1}{R^3} \propto T^2$ C. $R^2 \propto T^3$ D. $\frac{1}{R^2} \propto T^3$

$$T^2 = [4\pi^2 / (GM)] r^3$$

Kepler's third law

$$\cdot T^2 = [4\pi^2 / (GM)] R^3 \rightarrow R^3 = [GM / (4\pi^2)]^{-1} T^2.$$

· $R^3 \propto T^2$ with $1 / [GM / (4\pi^2)]$ being the constant of proportionality.

FYI

· Kepler's third law originally said that the square of the period was proportional to the cube of the radius – and nothing at all about what the constant of proportionality was. Newton's law of gravitation was needed for that!

Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving orbital period

The rings of Saturn are made of rocky particles that orbit the planet. The period T of each particle depends on its distance r from the centre of Saturn. The period T is proportional to r^n .

Which **one** of the following is n equal to?

- A. 1.0 **B. 1.5** C. 2.0 D. 3.0

$$T^2 = [4\pi^2 / (GM)] r^3$$

Kepler's third law

$$\cdot T^2 = [4\pi^2 / (GM)] r^3$$

$$T = \{ [4\pi^2 / (GM)] r^3 \}^{1/2}$$

$$T = [4\pi^2 / (GM)]^{1/2} r^{3/2}$$

$$\cdot \text{Thus } T \propto r^{3/2}.$$

Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving orbital period

Two satellites, X and Y, move in circular orbits about the Earth. The orbital period of satellite X is eight times that of satellite Y.

The ratio $\frac{\text{orbital radius of satellite X}}{\text{orbital radius of satellite Y}}$ is

- A. 2. **B. 4.** C. 8. D. 16.

$$T^2 = [4\pi^2 / (GM)] r^3$$

Kepler's third law

$$\cdot T_X^2 = [4\pi^2 / (GM)] r_X^3.$$

$$\cdot T_Y^2 = [4\pi^2 / (GM)] r_Y^3.$$

$$\cdot T_X = 8T_Y \rightarrow T_X^2 = 64T_Y^2.$$

$$T_X^2 / T_Y^2 = \cancel{[4\pi^2 / (GM)]} r_X^3 / \{ \cancel{[4\pi^2 / (GM)]} r_Y^3 \}$$

$$64 \cancel{T_Y^2} / \cancel{T_Y^2} = r_X^3 / r_Y^3$$

$$64 = (r_X / r_Y)^3$$

$$r_X / r_Y = 64^{1/3} = 4$$

Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational field

- Consider Dobson inside an elevator which is not moving...
- If he drops a ball, it will accelerate downward at 10 ms^{-2} as expected.



PRACTICE: If the elevator is accelerating upward at 2 ms^{-2} , what will Dobson observe the dropped ball's acceleration to be?

SOLUTION:

- Since the elevator is accelerating upward at 2 ms^{-2} to meet the ball which is accelerating downward at 10 ms^{-2} , Dobson would observe an acceleration of 12 ms^{-2} .
- If the elevator were accelerating downward at 2, he would observe an acceleration of 8 ms^{-2} .

Topic D - Fields

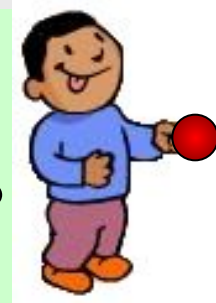
D.1 Gravitational fields (SL)

Solving problems involving gravitational field

PRACTICE: If the elevator were to accelerate downward at 10 ms^{-2} , what would Dobson observe the dropped ball's acceleration to be?

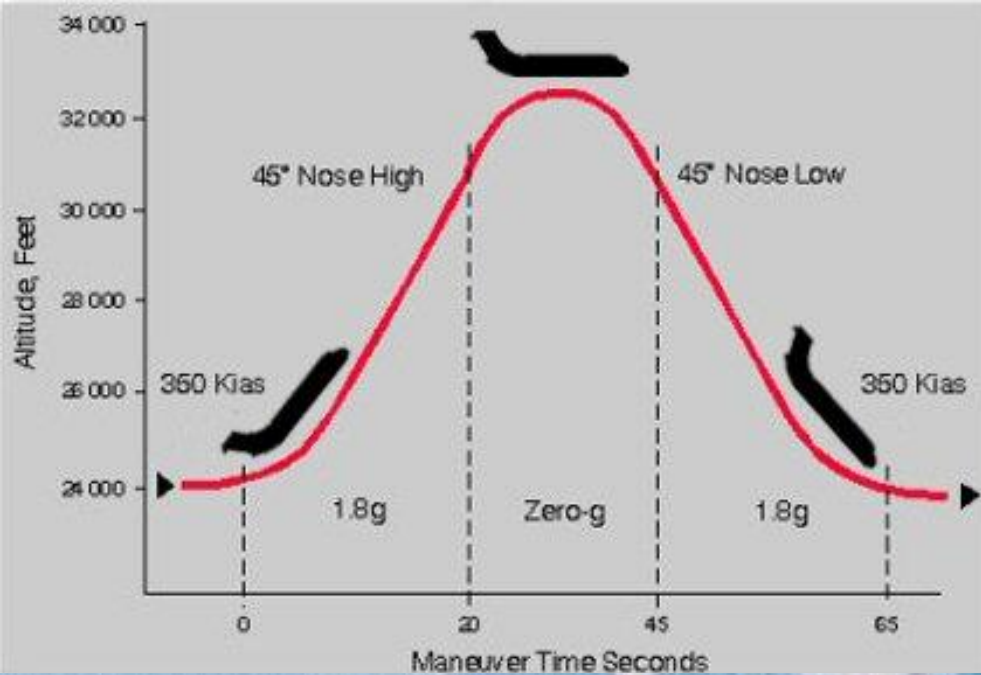
SOLUTION:

- He would observe the acceleration of the ball to be zero!
- He would think that the ball was “weightless!”



FYI

- The ball is NOT weightless, obviously. It is merely accelerating at the same rate as Dobson!
- How could you get Dobson to accelerate downward at 10 ms^{-2} ? **Cut the cable!**



The "Vomit Comet"

Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational field

PRACTICE: We have all seen astronauts experiencing “weightlessness.” Explain why it only appears that they are weightless.

SOLUTION: The astronaut, the spacecraft, *and* the tomatoes, are all accelerating at $a_c = g$.

· They all fall together and appear to be weightless.



Topic D - Fields

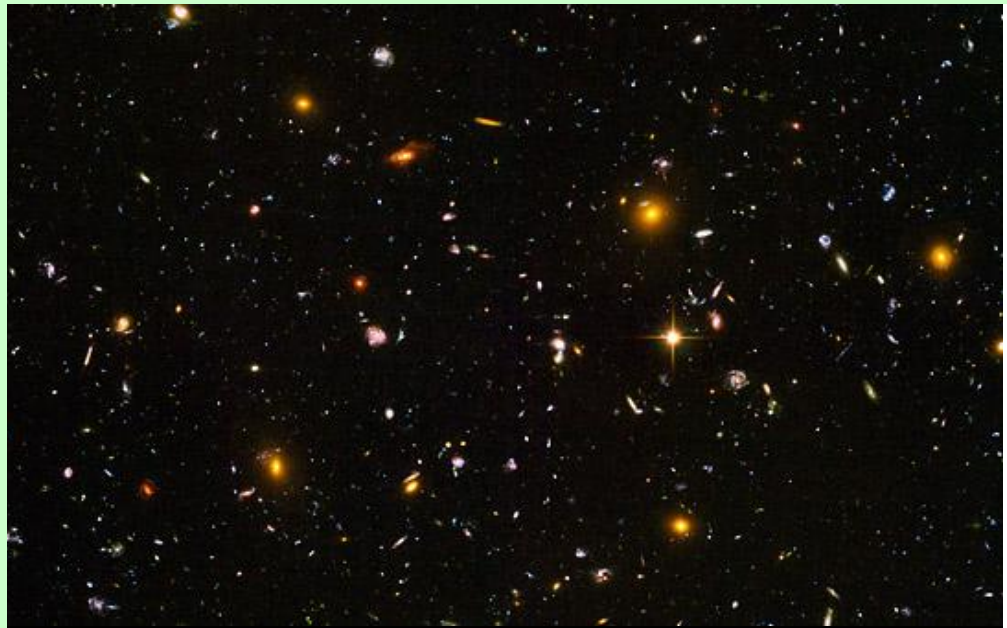
D.1 Gravitational fields (SL)

Solving problems involving gravitational field

PRACTICE: Discuss the concept of weightlessness in **deep space**.

SOLUTION: Only in deep space – which is defined to be far, far away from all masses – will a mass be truly weightless.

· In deep space, the r in $F = GMm / r^2$ is so large for every m that F , the force of gravity, is for all intents and purposes, zero.



Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational field

A satellite of mass m and speed v orbits the Earth at a distance r from the centre of the Earth. The gravitational field strength due to the Earth at the satellite is equal to

- A. $\frac{v}{r}$. **B. $\frac{v^2}{r}$.** C. $\frac{mv}{r}$. D. $\frac{mv^2}{r}$.

- Since the satellite is in circular orbit $F_c = mv^2/r$.
- Since the satellite's weight is holding it in orbit, $F_c = mg$.
- Thus ~~$mv^2/r = mg$~~ .
- Finally $g = v^2/r$.

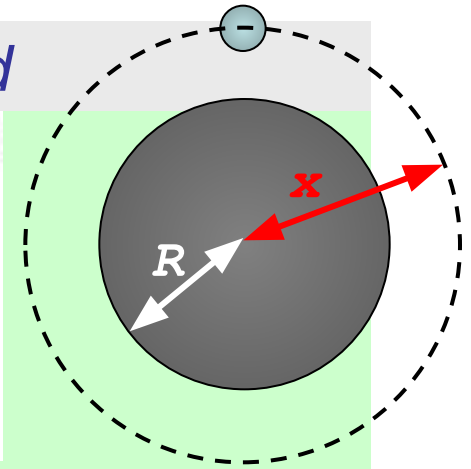
Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational field

A satellite of mass m orbits a planet of mass M and radius R as shown. (*The diagram is not to scale.*)

The radius of the circular orbit of the satellite is x . The planet may be assumed to behave as a point mass with its mass concentrated at its centre.



(a) Deduce that the linear speed v of the satellite in its orbit is

given by the expression $v = \sqrt{\frac{GM}{x}}$.

• $g = GM / x^2$

• $a_c = GM / x^2$ (since $a_c = g$ in circular orbits).

• $v^2 / x = GM / x^2$ (since $r = x$ and $a_c = v^2 / x$).

• $v^2 = GM / x$ so that $v = \sqrt{GM / x}$.

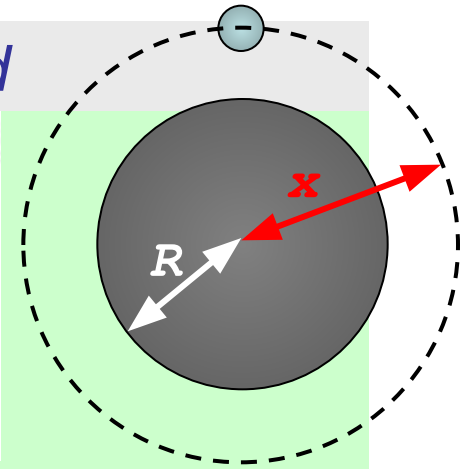
Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational field

A satellite of mass m orbits a planet of mass M and radius R as shown. (*The diagram is not to scale.*)

The radius of the circular orbit of the satellite is x . The planet may be assumed to behave as a point mass with its mass concentrated at its centre.



(b) Derive an expression, in terms of m , G , M and x , for the kinetic energy of the satellite.

• From (a) $v^2 = GM / x$.

• But $E_K = (1/2)mv^2$.

• Thus $E_K = (1/2)mv^2 = (1/2)m(GM / x) = GMm / (2x)$.

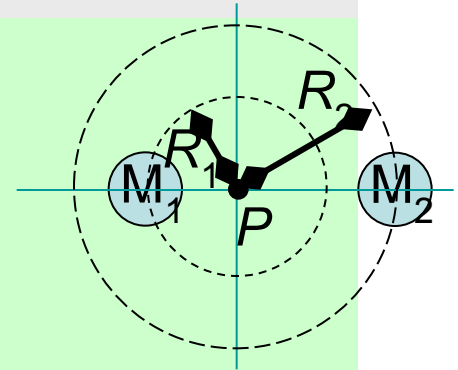
Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational force

A binary star consists of two stars that each follow circular orbits about a fixed point P as shown.

The stars have the same orbital period T . Each star may be considered to act as a point mass with its mass concentrated at its centre. The stars, of masses M_1 and M_2 , orbit at distances R_1 and R_2 respectively from point P .



- (a) State the name of the force that provides the centripetal force for the motion of the stars.

• It is the gravitational force.

.....

• Note that $F_g = GM_1M_2 / (R_1 + R_2)^2$.

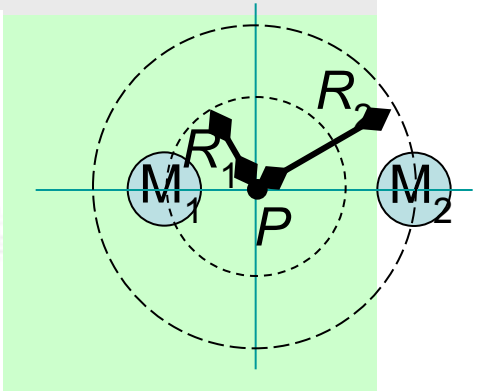
Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational force

A binary star consists of two stars that each follow circular orbits about a fixed point P as shown.

The stars have the same orbital period T . Each star may be considered to act as a point mass with its mass concentrated at its centre. The stars, of masses M_1 and M_2 , orbit at distances R_1 and R_2 respectively from point P.



(b) By considering the force acting on one of the stars, deduce that the

orbital period T is given by the expression $T^2 = \frac{4\pi^2}{GM_2} R_1 (R_1 + R_2)^2$.

• M_1 experiences $F_c = M_1 v_1^2 / R_1$.

• Since $v_1 = 2\pi R_1 / T$, then $v_1^2 = 4\pi^2 R_1^2 / T^2$.

• Thus $F_c = F_g \rightarrow M_1 v_1^2 / R_1 = GM_1 M_2 / (R_1 + R_2)^2$.

$$\cancel{M_1} (4\pi^2 R_1^2 / T^2) / \cancel{R_1} = \cancel{GM_1} M_2 / (R_1 + R_2)^2$$

$$4\pi^2 R_1 (R_1 + R_2)^2 = GM_2 T^2$$

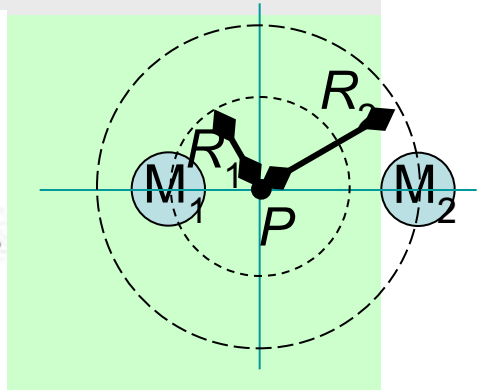
$$\cdot T^2 = \frac{4\pi^2}{GM_2} R_1 (R_1 + R_2)^2$$

Topic D - Fields

D.1 Gravitational fields (SL)

Solving problems involving gravitational force

A binary star consists of two stars that each follow circular orbits about a fixed point P as shown.



The stars have the same orbital period T . Each star may be considered to act as a point mass with its mass concentrated at its centre. The stars, of masses M_1 and M_2 , orbit at distances R_1 and R_2 respectively from point P.

- (c) The star of mass M_1 is closer to the point P than the star of mass M_2 . Using the answer in (b), state and explain which star has the larger mass.

• From (b) $T^2 = (4\pi^2 / GM_2)R_1(R_1 + R_2)^2$.

• From symmetry $T^2 = (4\pi^2 / GM_1)R_2(R_1 + R_2)^2$.

~~$(4\pi^2 / GM_2)R_1(R_1 + R_2)^2 = (4\pi^2 / GM_1)R_2(R_1 + R_2)^2$~~

$$(1 / M_2)R_1 = (1 / M_1)R_2$$

$$M_1 / M_2 = R_2 / R_1$$

• Since $R_2 > R_1$, we see that $M_1 > M_2$.